

SolarGrid Demo Utility — Generator Technical Requirements

SG-GEN-002

SOLARGRID AI — Synthetic Engineering Demonstration Corpus

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Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY
Purpose	Define synthetic technical and operational parameters for G1, G2, and G3.

Demo Note: This document contains synthetic engineering requirements created for the SOLARGRID AI demonstration. All numerical values, thresholds, equipment ratings, and procedures are illustrative and must be replaced or validated by the deploying organization before operational use.

Use restriction: This document is intended only for RAG testing, semantic retrieval testing, SafetyAgent demonstration, engineering evidence demonstration, PlannerAgent integration testing, and Tool 16 testing. It is not intended for real-world grid operation.

Document Control

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Revision History

Revision	Date	Description	Author
1.0	2026-01-15	Initial synthetic demonstration release	SolarGrid Demo Utility — Engineering Department

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1. Purpose and Scope

This manual defines synthetic technical requirements for three gas-fired generating units used in the SolarGrid Demo Utility demonstration network. It covers active-power limits, ramping, commitment constraints, startup behavior, reserve contribution, efficiency, fuel-cost assumptions, and maintenance availability.

The generator parameters are illustrative engineering data for software testing. They are not equipment manufacturer data, contractual terms, or regulatory requirements.

2. Generator Fleet Summary

Generator	Pmin	Pmax	Ramp Up	Ramp Down	Min Up	Min Down	Startup
G1	20 MW	100 MW	15 MW/min	20 MW/min	60 min	30 min	12 min
G2	15 MW	80 MW	12 MW/min	16 MW/min	45 min	30 min	10 min
G3	10 MW	60 MW	10 MW/min	12 MW/min	30 min	20 min	8 min

GEN-001 — G1 has a synthetic dispatch range of 20–100 MW.

GEN-002 — G2 has a synthetic dispatch range of 15–80 MW.

GEN-003 — G3 has a synthetic dispatch range of 10–60 MW.

3. Active Power and Ramping

GEN-004 — G1 shall not exceed 100 MW and shall not operate below 20 MW while synchronized and committed.

GEN-005 — G2 shall not exceed 80 MW and shall not operate below 15 MW while synchronized and committed.

GEN-006 — G3 shall not exceed 60 MW and shall not operate below 10 MW while synchronized and committed.

GEN-007 — G1 ramp-up capability is 15 MW/min and ramp-down capability is 20 MW/min.

GEN-008 — G2 ramp-up capability is 12 MW/min and ramp-down capability is 16 MW/min.

GEN-009 — G3 ramp-up capability is 10 MW/min and ramp-down capability is 12 MW/min.

Ramp constraints are evaluated over the requested dispatch interval. A planner may not assume instantaneous movement between two dispatch points.

4. Commitment and Minimum Up/Down Times

GEN-010 — G1 shall remain committed for at least 60 minutes after startup and shall remain off for at least 30 minutes after shutdown.

GEN-011 — G2 shall remain committed for at least 45 minutes after startup and shall remain off for at least 30 minutes after shutdown.

GEN-012 — G3 shall remain committed for at least 30 minutes after startup and shall remain off for at least 20 minutes after shutdown.

These values are synthetic commitment constraints intended to make unit commitment and replanning tests non-trivial.

5. Startup Constraints

GEN-013 — Synthetic startup lead times are 12 minutes for G1, 10 minutes for G2, and 8 minutes for G3.

GEN-014 — A generator shall not be credited as immediately available for reserve during a startup interval unless the operating plan explicitly models the required response and synchronization state.

6. Reserve Contribution

Generator	Maximum Synthetic Reserve Contribution	Constraint
G1	25 MW	Cannot exceed available headroom or stated reserve capability
G2	20 MW	Cannot exceed available headroom or stated reserve capability
G3	15 MW	Cannot exceed available headroom or stated reserve capability

GEN-015 — Generator reserve contribution shall be limited by both unit headroom and the synthetic reserve capability in the fleet table.

GEN-016 — A generator operating at Pmax has zero upward headroom and shall not be counted as upward spinning reserve.

7. Efficiency and Fuel Cost Assumptions

Generator	Synthetic Net Efficiency	Fuel Cost Assumption
G1	39%	62 currency units/MWh
G2	41%	58 currency units/MWh
G3	38%	66 currency units/MWh

GEN-017 — The listed efficiency and fuel-cost values are planning assumptions only and shall not be treated as measured plant performance.

8. Maintenance Constraints

GEN-018 — A generator marked unavailable for maintenance shall have zero dispatch and zero reserve contribution in the current plan.

GEN-019 — A maintenance availability change affecting one or more units shall trigger plan-impact assessment under the synthetic monitoring procedure.

Example maintenance calendar: G1 scheduled unavailable for 90 minutes during a synthetic low-load window; G2 and G3 remain available. The example is intentionally configurable.

9. Validation Examples

Example A: G1 moves from 50 MW to 80 MW over two minutes. The requested 30 MW change requires 15 MW/min and is consistent with the synthetic ramp-up limit.

Example B: G3 moves from 55 MW to 30 MW over one minute. The requested 25 MW reduction exceeds the 12 MW/min ramp-down limit and therefore requires a longer transition or a different dispatch sequence.

A numerical optimization solution is not sufficient evidence of generator feasibility unless unit limits, ramping, commitment state, and availability have been validated.